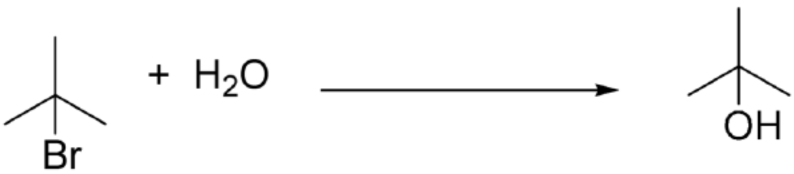
QXU4004 – Engineering Chemistry
Alcohols, Alkyl halides and Substitution

1) Why does the stability of carbocations follow the trend: tertiary > (more stable than) secondary > primary > methyl? Consider both hyperconjugation and electronic effects.

2) For the following reaction, identify the type of mechanism and draw it, and discuss what type of solvent you would use (discuss the properties of the solvent required, and propose specific solvents)



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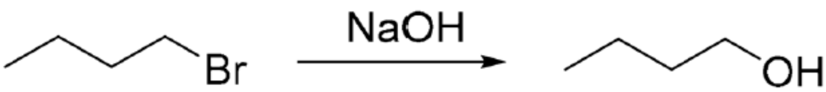

4) Consider the following reaction, which could proceed by either SN1 or SN2.

Discuss the effect on k_{SN1}/k_{SN2} when you change the following conditions:

a) X: Cl, I
b) Y: H_2O, NaOH
c) Solvent: Acetone, Aqueous Ethanol

5) Propose structures for A, B and C, and draw mechanisms for the reactions below. B and C are structural isomers. B is the major product of the second reaction.